CO543/CO5430
Computer Vision Project Assignment
2026

Item	Specification
Course component	The project contributes 25% of the final course mark.
Project period	1 July 2026 to 7 September 2026.
Group size	Up to 4 students per group. Individual contribution is assessed.
Project type	Practical Computer Vision project with implementation, experiments, evaluation, report, repository, and final demonstration.
Submission date	Final submission and demonstration: 7 September 2026.

1. Assignment Overview

This project is a half-semester group project in which students design, implement, evaluate, and demonstrate a practical Computer Vision solution. The project should move beyond basic image processing and must address a clear vision task such as recognition, detection, segmentation, tracking, matching, reconstruction, pose estimation, optical flow, localization, depth estimation, or another approved vision problem.

- **Primary aim:** give students hands-on experience with the complete Computer Vision workflow: problem formulation, dataset preparation, model/algorithm selection, implementation, evaluation, analysis, and communication.
- **Course weight:** 25% of the final course mark.
- **Expected level:** Clear baseline, measurable evaluation, and evidence of independent technical decisions.

2. Learning Outcomes

- Define a Computer Vision problem with a measurable objective and feasible scope.
- Prepare or use a dataset responsibly, including train/validation/test splits and appropriate annotation handling.
- Implement a suitable baseline and at least one improved/adapted method.
- Evaluate results using task-appropriate quantitative metrics and qualitative visual evidence.
- Analyze failure cases, limitations, data bias, computational constraints, and possible improvements.
- Maintain a reproducible GitHub repository with setup instructions and documented experiments.
- Communicate technical work clearly through progress reviews, a final report, and a live demonstration.

3. Scope and Acceptability Rules

3.1 Acceptable projects

- The project must use visual data: images, video, RGB-D, stereo images, point clouds generated from images, medical/satellite/industrial images, or other approved visual data.
- The project must include implementation and experimental evaluation, not only a literature review.
- A pretrained model may be used, but the group must adapt, fine-tune, evaluate, compare, or integrate it in a meaningful way.
- A classical Computer Vision project is acceptable if it includes algorithmic implementation, parameter study, and quantitative/qualitative evaluation.
- Each project must define a minimum goal, expected goal, and stretch goal.

3.2 Not acceptable

- Only applying simple image enhancement such as filtering, sharpening, denoising, histogram equalization, or geometric transformation without a vision objective.
- Only running an existing library or tutorial without modification, evaluation, or explanation.
- Only building a user interface or web app without a Computer Vision method and evaluation.
- Using a dataset or external code without citation, license checking, or understanding.
- Submitting work that group members cannot explain during the demonstration + viva.

4. Group Formation and Topic Selection

- Groups may contain up to 4 students.
- All group members must participate in progress reviews and final demonstration unless formally excused.
- Topics may be selected from the approved topic bank or proposed by students (**Approval form IIC**).
- Suggested topics are allocated on a first-come, first-served basis, subject to approval of IIC.
  - Link: https://docs.google.com/spreadsheets/d/1yhjPYn3CSMqJPAXZxV5Jg68pHStup6uy/edit?usp=sharing&ouid=113724070324946976968&rtpof=true&sd=true
- Different groups may work in the same broad area, but they must differ in dataset, problem setting, method, or implementation objective.
- Student-proposed topics are approved using the criteria below.

Approval criterion	What must be shown
Computer Vision objective	The problem must involve visual understanding, measurement, reconstruction, recognition, or decision-making from images/video.
Feasible scope	The project must be achievable between 1 July and 7 September with available computing resources.
Dataset availability	Dataset source, size, labels/annotations, license, and access method must be clear.
Evaluation plan	The group must state suitable metrics and at least one baseline for comparison.
Technical contribution	There must be more than a direct re-run of a tutorial: adaptation, fine-tuning, comparison, ablation, pipeline integration, or algorithmic development is required.
Ethics and privacy	Projects involving people, faces, medical images, surveillance, or sensitive data require extra care and may require topic restrictions.

5. Recommended Project Tracks

- **Application track:** apply Computer Vision methods to a real-world problem in a domain such as agriculture, transport, medical/biomedical images, manufacturing, education, environment, remote sensing, sports, or campus operations.
- **Model/method track:** compare, improve, or adapt a vision model or algorithm for a specific task. This can be classical, deep-learning based, or a hybrid approach.
- **Systems/demo track:** build an end-to-end working pipeline, for example real-time detection/tracking, stitching, pose-based interaction, or visual localization. The system must still include evaluation.

6. Suggested Project Areas and Example Ideas

Area	Example project	Possible datasets	Metrics	Expected contribution
Image classification	Plant disease or waste-category classification	PlantVillage, local collected images, Kaggle datasets	Accuracy, F1-score, confusion matrix	Fine-tune ResNet/MobileNet; compare augmentation settings.
Object detection	Helmet/PPE detection, traffic sign detection, pothole detection	Roboflow public sets, COCO subset, local annotated data	mAP, precision, recall	YOLO/SSD baseline; evaluate on unseen scenes.
Semantic segmentation	Road/lane/vegetation/building segmentation	CamVid, Cityscapes subset, custom drone/satellite images	IoU, Dice, pixel accuracy	U-Net/DeepLab or classical + deep comparison.
Object tracking	Vehicle/person/sports player tracking in video	MOT-style data, local video, public sports footage	MOTA/ID switches, tracking precision, qualitative paths	Detector + SORT/DeepSORT or optical-flow based tracking.
Pose/gesture	Hand gesture control or exercise pose assessment	MediaPipe landmarks, custom webcam videos	Accuracy, confusion matrix, latency	Landmark features + classifier; real-time demo.
Feature matching and RANSAC	Image stitching/panorama construction	Own overlapping photos, Oxford affine, HPatches	Inlier count, reprojection error, visual alignment	SIFT/ORB + RANSAC + blending.
Stereo depth	Depth map from stereo pair	Middlebury, KITTI subset	Bad-pixel rate, RMSE, visual depth maps	Block matching vs SGBM; parameter study.
Optical flow	Motion estimation in traffic or sports video	Sintel, KITTI flow, custom video	Endpoint error or qualitative flow	Farneback/RAFT comparison if feasible.

Area	Example project	Possible datasets	Metrics	Expected contribution
Monocular depth	Estimate depth from a single image	NYU Depth v2 subset, KITTI subset	AbsRel, RMSE, qualitative depth maps	Pretrained model + dataset-specific analysis.
Visual localization	Recognize place or align camera pose from images	Campus image set, Aachen/other small localization set	Localization accuracy, matching inliers	Feature matching + geometric verification.
Medical/biological vision	Cell/lesion/object segmentation or classification	Public biomedical datasets	Dice, IoU, sensitivity, specificity	Ethics/license review required. Use public de-identified data.
Remote sensing	Building/road/vehicle detection in aerial images	DOTA subset, SpaceNet, Sentinel/Google Earth snapshots with care	mAP, IoU, precision/recall	Handle scale and orientation variation.

7. Project Timeline and Milestones

The project runs from 1 July to 7 September 2026. Dates may be adjusted slightly to match the teaching timetable, but the final submission date is fixed unless formally changed by the course coordinator.

ID	Milestone	Date/period	Expected deliverable	Project marks
M0	Group formation and topic registration	1 Jul - 7 Jul	Group members, topic category, dataset source, preliminary GitHub link	Formative approval
M1	Proposal and project plan	14 Jul	1-2 page proposal: problem, related work, dataset, method, evaluation, timeline, risks	10
M2	Dataset and baseline checkpoint	28 Jul	5 slides: dataset status, preprocessing/annotation plan, baseline pipeline, initial run, issues	10
M3	Prototype and preliminary results checkpoint	18 Aug	5-7 slides: working method, preliminary metrics, qualitative outputs, failure cases, next steps	15
M4	Experiment freeze	25 Aug	Final experimental plan fixed; remaining work focused on tuning, analysis, report, demo	Formative
M5	Draft report	1 Sep	Report skeleton, result tables/figures, demo checklist, repository cleanup	Formative
M6	Final submission and demonstration	7 Sep	Final report, GitHub repository, presentation slides, demo/video if applicable	65

8. Required Deliverables

Deliverable	Required content
Project proposal	Problem statement, motivation, dataset, planned method, baseline, evaluation metrics, related work, risks, and timeline.
Progress review slides	Short slides for checkpoint discussion: current status, evidence of implementation, problems, and next actions. All members should speak.
GitHub repository	Clean source code, README, setup instructions, dataset instructions, trained-model handling, demo/inference script or notebook, requirements file, result examples, citations.
Final report (IEEE Format)	6-8 pages recommended: abstract, introduction, related work, data, method, experiments, results, analysis, limitations, conclusion, references, contribution statement.
Final presentation and demonstration	7-10 minute presentation/demo plus Q&A. Must show problem, method, quantitative results, qualitative results, failure cases, and a working demonstration if possible.
Individual contribution statement	One table listing member roles: data, implementation, experiments, report, demo, repository. Lack of contribution may affect individual marks.

9. Technical Requirements

- **Baseline required:** Every project must include a simple baseline. Examples: pretrained model without fine-tuning, classical method, simple CNN, ORB/SIFT baseline, simple thresholding baseline for segmentation, or a naive tracker.
- **Improved/adapted method required:** There must be at least one meaningful extension or adaptation beyond the baseline.
- **Data split:** Use train/validation/test split where applicable. For small datasets, use cross-validation or a clear held-out test set.
- **Evaluation:** Use task-appropriate metrics and qualitative examples. Report both successes and failures.

- **Ablation or comparison:** At least one comparison should be included, such as with/without augmentation, classical vs deep method, different threshold values, different model sizes, or different feature detectors.
- **Reproducibility:** Another student should be able to run the main experiment or demo using the README.
- **Resource awareness:** Keep model size, runtime, memory, and computing platform realistic for undergraduate work.

10. Evaluation Scheme

The project is marked out of 100 and scaled to 25% of the final course mark.

Assessment component	Marks	Criteria
Proposal and planning	10	Clear problem, motivation, dataset, method plan, evaluation plan, feasible timeline, risk awareness.
Checkpoint 1: dataset and baseline	10	Dataset prepared or access confirmed, baseline implemented or partly working, early evidence shown, next steps clear.
Checkpoint 2: prototype and preliminary results	15	Working pipeline, preliminary quantitative/qualitative results, analysis of difficulties, revised plan.
Technical implementation	20	Correctness, completeness, code quality, appropriate use of methods/frameworks, meaningful adaptation beyond tutorial.
Experiments and evaluation	15	Suitable metrics, train/validation/test discipline, comparisons, ablation/parameter study, credible result tables/figures.
Results analysis and limitations	10	Interprets results, failure cases, bias/limitations, explains why method works or fails.
GitHub repository and reproducibility	10	README, environment, code organization, instructions, citations, lightweight sample data/inference demo, clear commit history.
Final report, presentation, demo, and Q&A	10	Clear report, effective presentation, working demo or evidence, ability to answer questions.

11. Marking Guidance by Project Quality Level

Level	Typical evidence
Excellent	Clear problem, strong dataset handling, robust implementation, multiple experiments, meaningful comparisons/ablations, strong analysis, reproducible repository, confident explanation by all members.
Good	Working implementation, reasonable dataset/evaluation, one or two comparisons, acceptable analysis, repository can be run with minor effort.
Satisfactory	Basic working system and demonstration, limited evaluation, some missing analysis or reproducibility weaknesses.
Weak	Incomplete pipeline, unclear dataset/evaluation, mainly copied code, poor explanation, or no convincing results.
Unacceptable	No Computer Vision objective, no implementation, plagiarism, no working evidence, or inability to explain submitted work.

12. Report Format

- Recommended length: 6-8 pages, excluding references and appendices. IEEE format.
- Use a clear technical report or conference-paper style format.
- Include figures, diagrams, result tables, and qualitative examples.
- Do not include large code listings in the report; provide code through GitHub.
- References are required for datasets, papers, pretrained models, external code, tutorials, and AI-generated assistance if used.

Section	Expected content
Title and authors	Project title, group number, member names, registration numbers, course code.
Abstract	Problem, method, dataset, key result in <= 300 words.
Introduction	Motivation, problem definition, contributions.
Related work	At least 3 relevant technical references; compare to your approach.
Data	Source, size, labels, preprocessing, split, limitations, license/privacy concerns.
Method	Pipeline, algorithms/models, training, hyperparameters, implementation details.
Experiments	Metrics, baseline, comparison, ablation/parameter study, quantitative results.

Section	Expected content
Results and discussion	Qualitative outputs, failure cases, interpretation, limitations.
Conclusion	What was achieved, what was learned, future improvements.
References and contributions	Bibliography and individual contribution statement.

13. GitHub Repository Requirements

Required item	Details
README.md	Problem statement, setup, how to run training/inference/demo, dataset instructions, expected outputs.
Environment file	requirements.txt, environment.yml, or notebook setup cell.
Source code structure	Separate code/notebooks for data preparation, training, evaluation, and demo/inference.
Results folder	Example outputs, plots, confusion matrix, detection/segmentation examples, failure cases.
Dataset handling	Do not upload restricted datasets. Provide download instructions or small permitted samples.
Model checkpoints	Do not commit large checkpoints. Provide link or instructions if needed.
References and license	Cite datasets, external code, papers, pretrained models, and tools. Include license notes.
AI use declaration	If AI tools were used, state where and how. Students remain responsible for understanding and correctness.

14. Dataset, Ethics, and Safety Requirements

- Use public datasets where possible. If collecting data, obtain consent where people are involved and avoid collecting sensitive personal information.
- Do not publish face images, medical images, or private videos unless the dataset license/ethics conditions clearly allow it.
- Report potential dataset bias and failure modes, especially for projects involving humans, medical decisions, security, or surveillance.
- Do not present model predictions as reliable medical, legal, or safety-critical decisions.
- Respect dataset licenses and cite sources in the report and repository.

15. AI Tool Use Policy

- AI tools may be used for coding assistance, debugging, explaining errors, improving writing clarity, or generating boilerplate, but their use must be declared.
- AI-generated code must be reviewed, tested, and understood by the group.
- The final report must represent the group's own technical understanding. AI should not replace analysis, experimentation, or explanation.
- During evaluations, students may be asked to explain any part of the code, method, or report. Inability to explain submitted material may reduce marks.
- Include a short AI-use statement in the report and README, even if no AI tools were used.

16. Progress Review Format

- Each checkpoint discussion is approximately 8-10 minutes per group.
- Each group should prepare concise slides and show code/results where available.
- All members should be ready to explain their own contribution.
- The monitoring sheet should be updated after each review by the lecturer/TA.

Checkpoint	What students should show
Checkpoint 1	Final problem statement, dataset proof, preprocessing/annotation plan, baseline design, GitHub link, early code/output, risks.

Checkpoint	What students should show
Checkpoint 2	Working prototype, preliminary metrics, qualitative examples, comparison to baseline, failure cases, revised timeline.
Final demo	Final results, live or recorded demo, reproducibility evidence, report, repository, individual contributions.

17. Recommended Metrics by Task

Task	Recommended metrics
Classification	Accuracy, precision, recall, F1-score, confusion matrix, per-class performance.
Object detection	mAP, precision, recall, IoU threshold, FPS/runtime if real-time.
Segmentation	IoU, Dice coefficient, pixel accuracy, qualitative masks.
Tracking	Tracking precision, ID switches, MOTA/MOTP if using standard tracking evaluation, qualitative tracks.
Pose/gesture recognition	Classification accuracy/F1, keypoint error if ground truth available, latency.
Feature matching/stitching	Number of inliers, reprojection error, visual alignment quality, seam/blending quality.
Stereo/depth	RMSE, AbsRel, bad-pixel rate, qualitative depth maps.
Optical flow	Endpoint error if ground truth exists, qualitative flow visualization.
Localization/visual odometry	Pose/translation error, matching inliers, trajectory visualization.

18. Submission Checklist

- [ ] Project title, group number, and member details are correct.
- [ ] Report includes problem, dataset, method, experiments, results, analysis, limitations, references, and contribution statement.
- [ ] GitHub repository link is accessible to the lecturer/TA.
- [ ] README explains how to install and run the project.
- [ ] Dataset source, license, and preprocessing are documented.
- [ ] Baseline and improved/adapted method are both reported.
- [ ] Quantitative metrics and qualitative examples are included.
- [ ] Failure cases and limitations are discussed.
- [ ] External code, pretrained models, papers, datasets, and AI tool use are cited.
- [ ] Large datasets/checkpoints are not committed directly to GitHub.